F12 4th CAR

# Lecture 34: Section 5.3
# Solving Trigonometric Equations

Solving trigonometric equations

Trigonometric equations of quadratic types

Trigonometric functions of multiple angles

Using inverse trigonometric funtions

Reminder:

Period for sin, cos, csc, sec is $2\pi$.

Period for tan, cot is $\pi$.

$$\tan\frac{\pi}{6} = \frac{\sin\frac{\pi}{6}}{\cos\frac{\pi}{6}} = \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{1}{\cancel{2}}\cdot\frac{\cancel{2}}{\sqrt{3}} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$$

## Lecture 34: Section 5.3
## Solving Trigonometric Equations

1. Reduce the trigonometric equation to one of the form:

Look in

$$\sin\theta = c \qquad (-1 \le c \le 1) \quad [0, 2\pi)$$

$$\cos\theta = c \qquad (-1 \le c \le 1) \quad [0, 2\pi)$$

$$\tan\theta = c \qquad (-\infty, \infty) \quad [0, \pi)$$

2. Find the solution $\theta$ in one period and then find the general solutions by adding an integer number of periods.

<u>ex.</u> Solve the equation $2\sin x - 1 = 0$. Find all solutions.

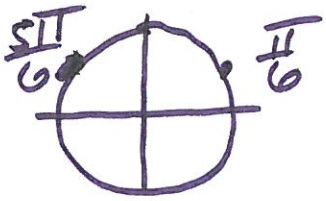
$2\sin x = 1$

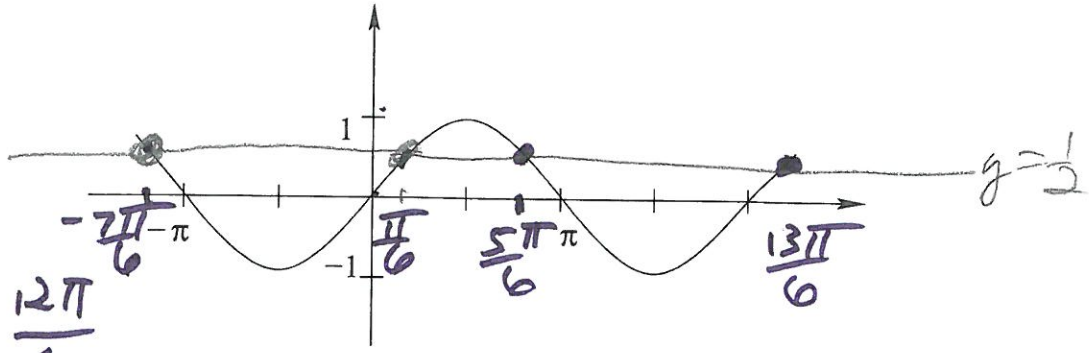
$\sin x = \frac{1}{2}$

General solutions:

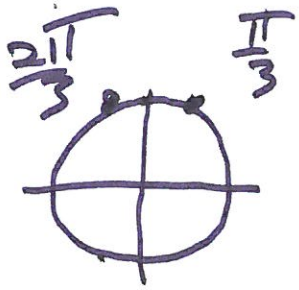
$x = \frac{\pi}{6} + 2n\pi = \frac{\pi}{6} + 2\pi n$

$x = \frac{5\pi}{6} + 2n\pi$ where $n$ is an integer




$2\pi = \frac{12\pi}{6}$




<u>ex.</u> Solve the equation $\tan^2 x - 3 = 0$.

$\tan^2 x = 3$

$\tan x = \pm\sqrt{3}$

$\tan x = \sqrt{3} \rightarrow x = \frac{\pi}{3} + n\pi$

or $\tan x = -\sqrt{3} \rightarrow x = \frac{2\pi}{3} + n\pi$

Get just one trig function.

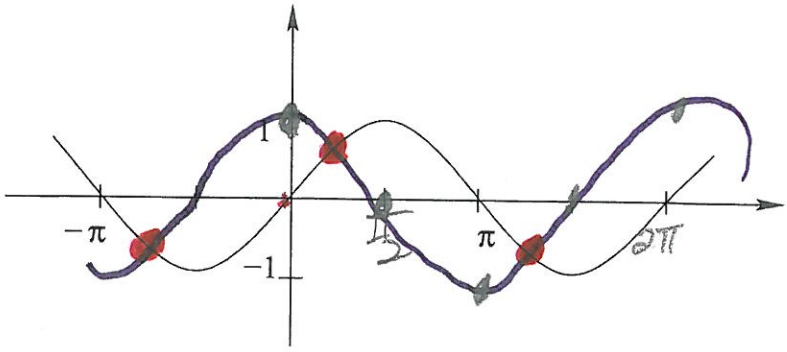
<u>ex.</u> Find the values of $x$ for which the graphs of $f(x) = \sin x$ and $g(x) = \cos x$ intersect.

$\sin x = \cos x$

thus $x = \frac{\pi}{4} + n\pi$

$\frac{\sin x}{\cos x} = \frac{\cos x}{\cos x}$

$\tan x = 1$



$\boxed{\begin{array}{l}\theta = 0 + 2n\pi \\ \theta = \pi + 2n\pi\end{array}}$ or $\boxed{\theta = n\pi}$

n is an integer

$\pi$ $0$

$\theta = 0 + n\pi = n\pi$

<u>ex.</u> Solve the equation $\sin^3\theta = 2\sin\theta$.

$\sin^3\theta - 2\sin\theta = 0$

$\sin\theta(\sin^2\theta - 2) = 0$

$\sin\theta = 0$ or $\sin^2\theta - 2 = 0$

$\sin^2\theta = 2$

$\sin\theta = \pm\sqrt{2}$ no solutions since $\sqrt{2} > 1$

**NOTE:** Do not cancel the factor $\sin\theta$ from both sides! This will eliminate some solutions. Don't divide by a variable if possible.

**Trigonometric Equations of Quadratic Types**

let $u = \sin x$

| Quadratic Equations | Quadratic Type Equations |
|---|---|
| $2x^2 - 3x + 1 = 0$ | $2\sin^2 x - 3\sin x + 1 = 0$ |
| $(2x-1)(x-1) = 0$ | $2u^2 - 3u + 1 = 0$ |
| $2x^2 - 5x + 2 = 0$ | $2\tan^2 x - 5\tan x + 2 = 0$ |
| | $(2u-1)(u-1) = 0$ |
| | $(2\sin x - 1)(\sin x - 1) = 0$ |

$$\boxed{x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}}$$

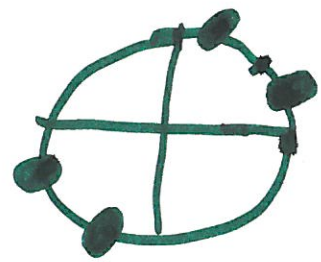
**ex.** Find all solutions in the interval $[0, 2\pi)$.

1) $2\sin^2 x - 3\sin x + 1 = 0$

$(2\sin x - 1)(\sin x - 1) = 0$

$2\sin x - 1 = 0$ or $\sin x - 1 = 0$

$\sin x = \frac{1}{2}$ $\qquad$ $\sin x = 1$



2) $2\tan^2 x - 5\tan x + 2 = 0$

$(2\tan x - 1)(\tan x - 2) = 0$

$2\tan x - 1 = 0$ or $\tan x - 2 = 0$

$\tan x = \frac{1}{2}$ $\qquad$ $\tan x = 2$

$$\boxed{\begin{aligned} &x = \arctan\tfrac{1}{2} \\ &x = \arctan\tfrac{1}{2} + \pi \\ &x = \arctan 2 \\ &x = \arctan 2 + \pi \end{aligned}}$$

3) $1 + \sin x = 2\cos^2 x$

$1 + \sin x = 2(1 - \sin^2 x)$

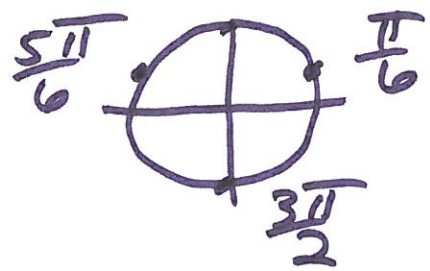
$1 + \sin x = 2 - 2\sin^2 x$

$2\sin^2 x + \sin x - 1 = 0$

$(2\sin x - 1)(\sin x + 1) = 0$

$2\sin x - 1 = 0$ or $\sin x + 1 = 0$

$\sin x = \frac{1}{2}$ $\qquad$ $\sin x = -1$




$$\boxed{x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}}$$

* Since we squared both side, check for extraneous solutions.

$=1$

**ex.** Find all solutions in the interval $[0, 2\pi)$.

1) $(\sin\theta + \cos\theta)^2 = (1)^2$

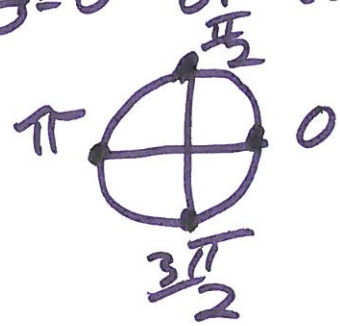
$\sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta = 1$

$1 + 2\sin\theta\cos\theta = 1$

$2\sin\theta\cos\theta = 0$

$\sin\theta\cos\theta = 0$

$\sin\theta = 0$ or $\cos\theta = 0$




Check: $\boxed{x=0}$ $\sin 0 + \cos 0 = 1$

$0 + 1 = 1$ ✓

$\boxed{x=\frac{\pi}{2}}$ $\sin\frac{\pi}{2} + \cos\frac{\pi}{2} = 1$

$1 + 0 = 1$ ✓

$\cancel{x=\pi}$ $\sin\pi + \cos\pi = 1$

$0 + (-1) \neq 1$

$\cancel{x=\frac{3\pi}{2}}$ $\sin\frac{3\pi}{2} + \cos\frac{3\pi}{2} = 1$

$-1 + 0 \neq 1$

**NOTE:** If we perform an operation on an equation that may introduce new roots, such as squaring both sides, then we must **check** that the solutions obtained are not extraneous.

Write in terms of sine and cosine

$$\frac{\sin\theta}{\cos\theta} + \frac{1}{\cos\theta} = 1$$

*watch for domain restrictions

2) $\tan\theta + \sec\theta = 1$

$$\left(\frac{\sin\theta + 1}{\cos\theta} = 1\right)\cos\theta$$

$$\sin\theta + 1 = \cos\theta$$

$$(\sin\theta + 1)^2 = (\cos\theta)^2$$

watch for extraneous sol.

$$\sin^2\theta + 2\sin\theta + 1 = \cos^2\theta$$

$$\sin^2\theta + 2\sin\theta + 1 = 1 - \sin^2\theta$$

$$2\sin^2\theta + 2\sin\theta = 0$$

$$\sin^2\theta + \sin\theta = 0$$

$$\sin\theta(\sin\theta + 1) = 0$$

$\sin\theta = 0$ or $\sin\theta + 1 = 0$

$\sin\theta = -1$

Check $\boxed{x = 0}$ $\tan 0 + \sec 0 = 1$

$0 + \frac{1}{1} = 1$ ✓

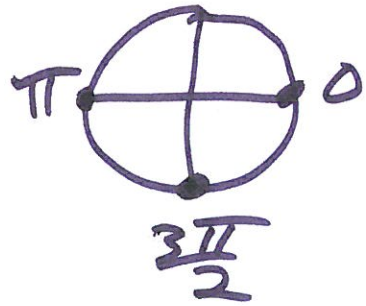
$\cancel{x = \pi}$ $\tan\pi + \sec\pi = 1$

$0 + -1 \neq 1$

$\cancel{x = \frac{3\pi}{2}}$ $\tan\frac{3\pi}{2} + \sec\frac{3\pi}{2} = 1$

$\frac{-1}{0} +$

undefined thus $\frac{3\pi}{2}$ is not in the domain



## Trigonometric Functions of Multiple Angles

When solving trigonometric equations that involve functions of multiples of angles, we first solve for the multiple of the angle, then divide to solve for the angle.

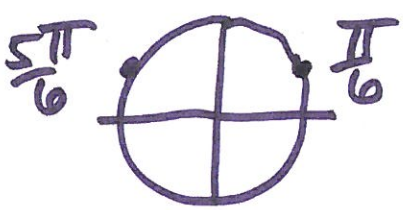
<u>ex.</u> Consider the equation $2\sin 3x - 1 = 0$.

$2\sin 3x = 1$

1) Find all solutions of the equation.

$\sin 3x = \frac{1}{2}$

what angles have a sine value of $\frac{1}{2}$?




Thus $3x = \frac{\pi}{6} + 2n\pi$

$$\boxed{x = \frac{\pi}{18} + \frac{2n\pi}{3}}$$

also $3x = \frac{5\pi}{6} + 2n\pi$

$$\boxed{x = \frac{5\pi}{18} + \frac{2n\pi}{3}}$$

2) Find the solutions in the interval $[0, 2\pi)$.

$$x = \frac{\pi}{18} + \frac{2n}{3}\pi = \frac{\pi}{18} + \frac{2\pi}{3}n$$

$n=0$ $\quad x = \frac{\pi}{18} + \frac{2\pi}{3}(0) = \frac{\pi}{18} + 0 = \boxed{\frac{\pi}{18}}$

$n=1$ $\quad x = \frac{\pi}{18} + \frac{2\pi}{3}(1) = \frac{\pi}{18} + \frac{12\pi}{18} = \boxed{\frac{13\pi}{18}}$

$n=2$ $\quad x = \frac{\pi}{18} + \frac{2\pi}{3}(2) = \frac{\pi}{18} + \frac{24\pi}{18} = \boxed{\frac{25\pi}{18}}$

$n=3$

$$= \cancel{\frac{37\pi}{18}} > 2\pi$$

Also

$$= \frac{5\pi}{18} + \frac{2\pi}{3}n$$

$n=-1$ $\quad x = \frac{5\pi}{18} + \frac{2\pi}{3}(-1) = \frac{5\pi}{18} + \frac{-2\pi}{3}$

$$= \frac{5\pi}{18} + -\frac{12\pi}{18} = -$$

$n=0$ $\quad x = \frac{5\pi}{18}$

$n=1$ $\quad x = \frac{17\pi}{18}$

$n=2$ $\quad x = \frac{29\pi}{18}$

$n=3$ $\quad x = \cancel{\frac{41\pi}{18}} > 2$

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**ex.** Consider the equation $\sqrt{3}\tan\frac{x}{2} - 1 = 0$.

1) Find all solutions of the equation.

$\sqrt{3}\tan\frac{x}{2} = 1$

$\tan\frac{x}{2} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

$\frac{x}{2} = \frac{\pi}{6} + 2n\pi$

also

$\frac{x}{2} = \frac{7\pi}{6} + 2n\pi$

$2\left(\frac{x}{2}\right) = \left(\frac{\pi}{6} + n\pi\right)2 \rightarrow \boxed{x = \frac{\pi}{3} + 2n\pi}$

where n is an integer.

2) Find the solutions in the interval $[0, 4\pi)$.

$x = \frac{\pi}{3} + 2n\pi$

$n=0 \quad x = \frac{\pi}{3} + 2(0\pi) = \frac{\pi}{3}$

$n=1 \quad x = \frac{\pi}{3} + 2(1)\pi = \frac{\pi}{3} + \frac{6\pi}{3} = \frac{7\pi}{3}$

$n=2 \quad x = \frac{\pi}{3} + 2(2)\pi = \frac{\pi}{3} + 4\pi =$ too big

## Using Inverse Trigonometric Functions

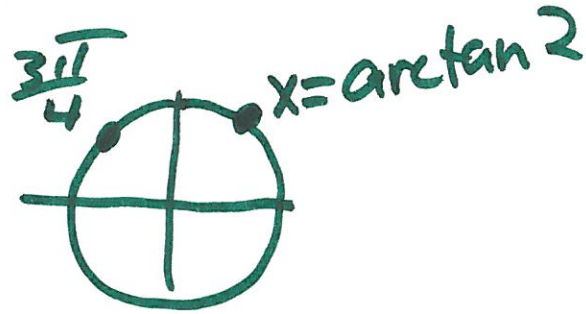
<u>ex.</u> Solve the equation $\sec^2 x - \tan x - 3 = 0$.

$1 + \tan^2 x - \tan x - 3 = 0$

$\tan^2 x - \tan x - 2 = 0$

$(\tan x - 2)(\tan x + 1) = 0$

$\tan x = 2 \qquad \tan x = -1$




$\tan \rightarrow$ check in $[0, \pi)$

General solutions: $\boxed{x = \arctan(2) + n\pi \qquad x = \frac{3\pi}{4} + n\pi}$